

EEE4114F Class Test 2

May 2024

Duration: 1 hour

Total: 40 marks

Empl ID: _____

Instructions:

- This is a closed-book exam.
- There are two parts to this exam.
- **Part A** has 2 questions totalling 20 marks. You must answer all of them.
- **Part B** has 2 questions totalling 20 marks. You must answer all of them.
- Both parts can be answered in the space provided on the exam script, and you can write on the back of the printed pages if required. Answer books will be available should you need additional writing space.
- You must use a pen to complete the exam — pencil will only be marked for drawings. Values indicated on drawings must be written in pen. Please show all of your working clearly — method and approach matter.
- A table of standard Fourier transform and z-transform pairs is provided separately
- You have 1 hour.

Question	Marks	Grade
1	10	
2	10	
3	10	
4	10	
Total:	40	

Part A: Digital Signal Processing

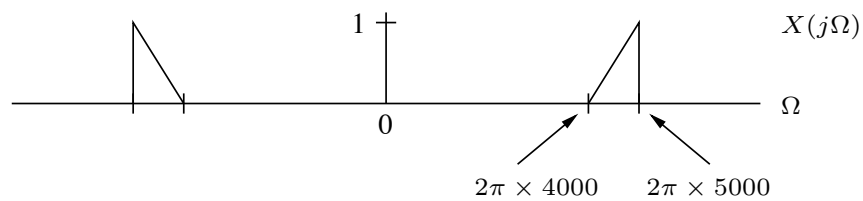
Question 1 [10 marks]

A system has impulse response $h[n] = (1/2)^n u[n]$. Determine the input $x[n]$ to the system if the output is given by

$$y[n] = \frac{1}{3}u[n] + \frac{2}{3}\left(-\frac{1}{2}\right)^n u[n].$$

Question 2 [10 marks]

A continuous-time signal $x_a(t)$ has the following Fourier transform:



1. What is the highest frequency present in the signal?
2. What is the lowest sampling frequency that can be applied to the signal such that no aliasing occurs and it can still be reconstructed? Draw the DTFT of $x[n]$ in this case.
3. Draw the block diagram of a reconstruction system that will recover $x_a(t)$ from the sampled signal $x[n]$ if the sampling rate is set to its lowest possible value. Sketch the frequency response of the reconstruction filter to use in this case.

Part B: Machine Learning

Question 3 [10 marks]

Suppose you train three different models: A, B, and C, as part of a task to predict the output power of a wind turbine based on wind speed. A plot of each model's predictions compared to the training data collected from a wind turbine is shown in Figure 1.

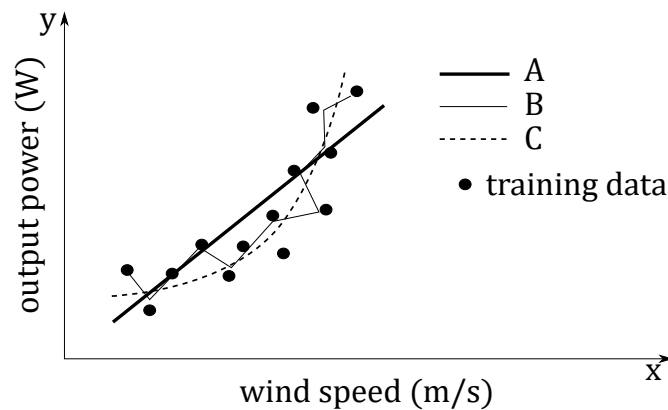


Figure 1: Plot of model predictions compared to training data

1. Is there sufficient information in Figure 1 to determine if model C would work well if deployed in a real-world wind farm? (2)

2. Figure 2 shows a plot of model capacity vs error. Which points on this plot (points indicated as 1, 2, or 3 on the x-axis) would you expect to correspond to each model (A, B, and C)? Explain your decisions and discuss the performance of each model with respect to bias and variance. (6)

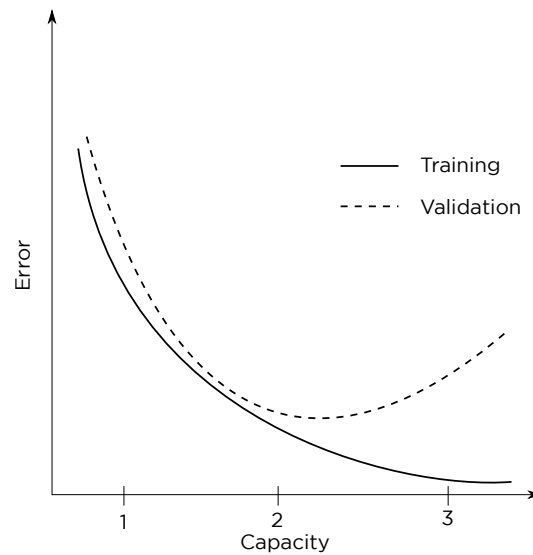


Figure 2: Model capacity vs. error

3. What could be done to improve model B without changing the number of model parameters and without adding additional training data? (2)

Question 4 [10 marks]

Suppose you need to develop a system that estimates the number of fruits a tree will produce based on the concentration of Potassium in the soil collected by a sensor buried in the ground. You collect a data set, $\mathcal{D}$, consisting of input-output pairs (x_i, y_i) . x_i is the concentration of Potassium measured in the soil for a single tree. y_i is the target variable, which, in this case, is the number of fruits collected from that tree.

$$\mathcal{D} = \{(1.1, 10), (2.0, 20), (1.4, 15), (2.8, 30)\}$$

You then decide to fit a regression model of the form, $h_\theta(x) = \theta_0 + \theta_1 x^2$, to it using gradient descent optimization methods. The objective function, J , you choose to optimize is given by the mean square error of the form:

$$J = \frac{1}{2N} \sum_{i=1}^N (h_\theta(x_i) - y_i)^2$$

Where N is the number of examples in a single training iteration.

The model's parameters are randomly initialised such that

$$\theta_0 = 0, \theta_1 = 1$$

Suppose you decide to divide the data set into two, equally sized mini-batches, keeping the order as it is such that

$$\mathcal{D}_1 = \{(1.1, 10), (2.0, 20)\}, \mathcal{D}_2 = \{(1.4, 15), (2.8, 30)\}$$

A single training iteration uses one mini-batch and the model parameters are updated after **each** iteration. The update rule is given by:

$$\theta_i := \theta_i - \eta \frac{\partial J}{\partial \theta_i}$$

Where $\eta = 0.1$. Answer the following questions:

1. Write down the expressions for the gradients of the objective function with respect to the model's parameters $\frac{\partial J}{\partial \theta_0}$ and $\frac{\partial J}{\partial \theta_1}$. (2)

2. Calculate the gradients of the parameters after one training iteration using the first mini-batch, $\mathcal{D}_1$. (2)

3. Calculate what the model would be parameters after the **second** training iteration.

(4)

4. Is the choice of hypothesis function appropriate for the data? Would this result in high or low bias?

(2)